

Project Initialization and Planning Phase

Date	06 July 2026
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Propose Solution) report

Project Overview	
objective	Develop an AI and Machine Learning-based flood prediction system that analyzes rainfall, river water levels, weather conditions, and historical flood data to predict flood risk accurately and provide early warnings.
scope	The project focuses on collecting flood-related data, training machine learning models, predicting flood risk levels, and displaying results through a user-friendly web application for disaster preparedness.
Problem statement	
Description	Floods cause significant damage to lives, infrastructure, agriculture, and the economy. Traditional forecasting methods often fail to provide timely and accurate predictions.
Impact	An accurate flood prediction system can: • Reduce loss of life and property. • Improve disaster preparedness. • Support government agencies in emergency planning. • Enable early evacuation and resource management.
Proposed solution	
Approach	Use Machine Learning algorithms to analyze historical flood data, rainfall, river levels, humidity, and weather information to predict the probability of flooding.
Key features	• Flood risk prediction using Machine Learning. • Real-time weather and rainfall analysis. • Early warning alerts.

	• Interactive dashboard with prediction results.
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Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel i5/Ryzen 5 or higher
Memory	RAM specifications	8 GB
Storage	Disk space for data models, and logs	256 GB or higher
Software		
Frameworks	Python frameworks	
Libraries	Additional libraries	Pandas, Numpy, scikit-learn, Matplotlib, Joblib
Development Environment	IDE	VS code / Jupyter Notebook
Data		
Data	Source , size , format	Kaggle flood prediction Dataset